NITHIN DHARAVATHU

Agentic AI Engineer | Python, FastAPI, LLMs & RAG

Hyderabad, India • +91 6304801278 • nithindharavath01@gmail.com • LinkedIn • GitHub • Portfolio

PROFESSIONAL SUMMARY

AI & ML graduate and Python backend developer (FastAPI, Flask) building Agentic AI applications with LLMs, RAG pipelines, semantic search, and LangChain/LangGraph agents. Built an end-to-end AI-powered incident management platform with vector search (Qdrant), tool-calling agents, and Dockerized CI/CD, and integrated ML models into production-style REST APIs during Microsoft and EY internships. Seeking an Agentic AI Engineer role to build intelligent, production-ready AI applications.

TECHNICAL SKILLS

Agentic AI: LLM APIs (OpenAI-compatible, Ollama), RAG, Semantic Search, Qdrant, HuggingFace Embeddings, AI Agents, Prompt Engineering, Tool Calling, HITL, MCP.
AI Frameworks: LangChain, LangGraph, HuggingFace, ML fundamentals.
Backend & APIs: Python, FastAPI, REST APIs, Pydantic, SQLAlchemy, JWT/OAuth2, RBAC.
Databases: PostgreSQL, MySQL, SQLite, Qdrant (Vector DB).
Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions (CI/CD), Railway, Vercel.
Tools & Practices: Git, GitHub, Pytest, n8n, SDLC, Debugging & Performance Optimization.
Languages: Python, JavaScript, SQL.

PROJECTS

AI-Powered Incident & Management Support System — Personal Project
Python, FastAPI, PostgreSQL, LangChain, LangGraph, Qdrant, HuggingFace

- Designed and built a backend AI platform to report, track, and resolve IT incidents, with LLM-powered classification of category, severity, priority, and assigned team, plus AI-generated incident summaries.
- Developed a LangGraph support agent with tool calling and human-in-the-loop review, with AI root-cause analysis.
- Containerized with Docker, automated testing and deployment through GitHub Actions CI/CD, with Pytest coverage.

Online Voting System — Personal Project
Python, FastAPI, MySQL, JWT, Jinja2

- Built a secure voting platform with JWT authentication, RBAC, and CSRF protection for Admin and Student roles.
- Developed 10+ backend modules (election management, vote verification, result publication, one-vote-per-user enforcement) on a normalized MySQL schema with 6+ relational tables, secure file uploads, and audit verification.

EXPERIENCE

Freelance Web Developer — 2026 – Present

- Build AI-integrated websites and online booking systems for local businesses using Python backends, REST APIs, and modern web technologies.
- Handle requirements gathering, development, deployment, and client communication end to end.

AI Azure Virtual Intern, Microsoft — May 2025 – June 2025

- Integrated Random Forest and XGBoost models into a FastAPI backend, exposing secure REST APIs for real-time insurance premium prediction.
- Validated model performance with ML metrics and added Pydantic validation and exception handling, improving prediction reliability and API efficiency.

NextGen Intern, EY (Spendly – Personal Expense Tracker) — Feb 2025 – March 2025

- Developed a Flask application (HTML/CSS/JS) with REST API integration, CRUD operations, authentication, and session handling using clean application architecture.
- Managed relational data with SQLite and SQLAlchemy in a database-driven application.

EDUCATION

B.Tech, Computer Engineering (AI & ML) – Vaagdevi College of Engineering, Warangal — 2026
CGPA/Percentage: 74.6% • Languages: English, Hindi, Telugu